First Steps with AI Tools

A practical introduction for teachers

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Introduction

Who I am

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What you'll get out of today

1. How much should I trust what AI gives me?
2. Why does AI work well for some teachers and not for others?
3. What do I do when my students start using AI?

Outline

1. Generative AI in plain language
2. Prompting for reliable results
3. AI for classroom content
4. Students using AI in assignments

Quick calibration

- Who has heard of ChatGPT?
- Who has tried it?
- Who has used it for work?

Generative AI in plain language

Generative AI in plain language

- **What it actually is**
- What it's good at
- Where it makes mistakes, and why

Prompting for reliable results

AI for classroom content

Students using AI in assignments

The core idea

Very sophisticated autocomplete.

Given some text, the AI predicts the next word, then the next, then the next.

Trained on huge amounts of text

These tools learned by reading. From all that text, the model picked up patterns of how language is used. Not facts.

Books

Articles & papers

Web pages & code

Trillions of words.

Trained on the internet, with all its flaws

You wouldn't trust every webpage.

Don't trust every AI answer.

The major tools

ChatGPT	**Claude**	**Gemini**	**Copilot**
OpenAI	Anthropic	Google	Microsoft

Different makers, similar core capability. Pick one, get familiar with it, then try others.

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Drafting and rephrasing

Drafting

"Draft a parent email explaining why we're starting a new unit on cell biology."

Rephrasing

"Rewrite this paragraph for a grade 8 reading level, keeping the same meaning."

Summarizing and explaining

Summarizing

"Summarize this 30-page chapter in 200 words a student can use as a study aid."

Explaining

"Explain photosynthesis to a 7th grader."

Brainstorming and generating variations

Brainstorming

"Give me ten hook ideas for a lesson on the causes of World War One."

Variations

"Take this exit-ticket question and write five versions of varying difficulty."

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Does the following sentence sound right?

"Marie Curie won her second Nobel Prize in physics in 1909 for her work on x-ray crystallography."

Hallucinations

The AI states things that sound true but aren't.

> *"Marie Curie won her second Nobel Prize in physics in 1909 for her work on x-ray crystallography."*

Confident. Detailed. Wrong on three counts.

Wrong year, wrong field, and Marie Curie did not work on x-ray crystallography.

Common mistakes AI makes

Math and counting

Arithmetic on big numbers, counting words in a paragraph.

Recent events

Without web search, the model's knowledge has a cutoff date.

Fabricated citations

Plausible authors, plausible titles, made-up references.

Specialized topics

Subtle errors in your specific subject that only an expert spots.

Different shapes, same root cause. Next slide.

Why it makes mistakes

It's predicting

plausible text.

"What word usually comes next?"

It's not retrieving

verified facts.

"Is this actually true?"

Plausible text and true text usually overlap. Sometimes they don't.

Every mistake on the previous slide is a version of that gap.

Mistakes are predictable

If you know how the AI works, you can predict where it'll let you down.

Lots of similar text in its training? Usually fine.

Common explanations, well-known concepts, standard phrasing.

Needs a precise fact, a date, a number, a quote? Verify.

Specific claims are exactly where plausible-text-prediction falls short.

Don't be discouraged

A list of common mistakes can feel like a list of reasons not to try.

It's a tool. You learn to use it.

Everything from here on is how to work with these limits, not against them.

What we covered

Generative AI in plain language

AI tools are sophisticated autocomplete trained on huge text.

They're great at drafting, rephrasing, summarizing, brainstorming.

They make mistakes when plausible text is different from true text.

Prompting for reliable results

Generative AI in plain language

Prompting for reliable results

- **Anatomy of a good prompt**
- Live demo
- Iterating and quick wins
- Knowing your tool

AI for classroom content

Students using AI in assignments

From "question" to "brief"

Asking

"Give me a quiz on photosynthesis."

→

Briefing

Who you are, what you need, who it's for, and what good looks like.

Treat the model like a smart but new colleague. Tell it everything it needs to know.

Six components of a good prompt

1 Role

2 Context

3 Task

4 Constraints

5 Format

6 Examples

Role, context, task

1	**Role**	Who the model should be. *"You are a high school biology teacher."*
2	**Context**	What it needs to know about your situation. *"Your class of 9th graders just covered the light reactions."*
3	**Task**	What you actually want it to do. *"Write three exit-ticket questions on the Calvin cycle."*

Constraints, format, examples

4	**Constraints**	Limits, audience, things to avoid. *"Keep vocabulary at a 9th grade level. No equations."*
5	**Format**	How the answer should be structured. *"Three multiple-choice questions, four options each, answer key at the end."*
6	**Examples**	One sample of what you want, when words aren't enough. *"Like this: [paste a question you've used before]."*

Weak prompt, strong prompt

Weak

"Make a worksheet about the French Revolution."

No role, no context, no audience.

Strong

You are a high school history teacher.

You have 25 grade 10 students who just read about the storming of the Bastille.

Write a 20-minute worksheet that builds analysis skills.

Use 10th grade vocabulary. Avoid graphic violence.

Format: 1 short source excerpt, 4 questions (2 recall, 2 analysis), answer key.

Five out of six components, in plain language.

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Students using AI in assignments

The task

Build a reading comprehension exercise on photosynthesis.

Grade 9 biology

What changed across the three versions

Simple	A general response. Not bad, but not aimed at our students.
Better	Adding context and audience tightened the output. Reading level matched.
Well-structured	Adding format and constraints gave us something we could use Monday.

The model didn't get smarter. The brief got clearer.

It's still predicting plausible text. Better prompts aim that prediction at what you want.

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Students using AI in assignments

Prompting as conversation

You don't have to get it right on the first try.

Read what the model gives you. Tell it what's off. Ask again.

Most good outputs are the third or fourth message, not the first.

Giving feedback in practice

- *"Make it shorter."*
- *"Add three harder questions."*
- *"Cut the last paragraph."*
- *"Use simpler vocabulary."*
- *"Try again, but for ESL students."*
- *"Give me three different versions."*

Short, plain corrections beat re-typing the whole prompt.

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AI for classroom content

Students using AI in assignments

Free tier and paid tier

Free tier

Where most teachers should start.

Capable enough for most teaching tasks.

Daily message limits. Plan for a quick chat, not all-day use.

Often a slightly older/weaker model than the paid tier.

Paid tier

Worth it only if you're using it daily.

Useful for very complex tasks.

Higher limits and longer answers. Able to iterate longer for higher quality outputs.

The current top-end model.

If a free-tier answer disappoints, the next move is a better prompt, not a credit card.

Additional modes

Web search

Lets the model look things up. Use it for current events, recent data, or anything where citations matter.

Thinking

Trades speed for better answers on hard problems. Worth turning on for complex reasoning, math, or multi-step tasks.

Persistent context

Memories

The model remembers things you've told it before, across chats.

Projects

A folder for related chats that share files and instructions.

Custom instructions

A standing brief you write once. Applies to every new chat.

These names and features change every few months. Learn to read what your tool offers.

Three reusable patterns

1. **The Brief** — *Default starter for any new task.* Role + task + constraints + format.

2. **The Critic** — *When you've already drafted something.* Paste your work, give criteria, ask for specific improvements.

3. **The Differentiator** — *When one version isn't enough.* Take any output, ask for it again at a different level or for a different student.

What we covered

Prompting for reliable results

Stop asking. Start briefing. Six components: role, context, task, constraints, format, examples.

The model didn't get smarter. The brief got clearer.

Iterate in plain language. Most good outputs are the third or fourth message.

Tools differ. Free vs paid, modes, and persistent context all matter.

AI for classroom content

Generative AI in plain language

Prompting for reliable results

AI for classroom content

- **Lesson materials**
- Student practice
- Formative assessment
- The verification habit

Students using AI in assignments

What AI helps you produce

Explanations

Multiple ways to explain the same concept until one lands.

Analogies

Bridges from familiar territory to unfamiliar ideas.

Worked examples

Step-by-step solutions you can adapt for class or homework.

Differentiated versions

The same content rewritten for different reading levels.

One topic, three reading levels

Topic: mitosis, grade 9 biology

Foundational

Below grade level

Cells make new cells by splitting in two. Before splitting, the cell makes a copy of its instructions, so each new cell gets a full set.

Grade level

Standard grade 9

In mitosis, a cell divides once to produce two genetically identical daughter cells. The DNA is replicated, the chromosomes line up, then separate, and the cell splits.

Advanced

Stretch readers

Mitosis proceeds through prophase, metaphase, anaphase, and telophase. Sister chromatids align at the metaphase plate before spindle fibers pull them to opposite poles.

Three ways to teach the same idea

Concept: how electrical resistance works

Explanation	Resistance is the opposition a material offers to electric current. More resistance, less current at the same voltage.
Analogy	Think of electrons flowing through a wire like water flowing through a pipe. A narrow pipe slows the water down. A high-resistance wire slows the electrons down.
Worked example	A 12 V battery is connected to a 4 ohm resistor. Using $V = IR$, the current is $I = 12 / 4 = 3$ amperes.

Generative AI in plain language

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Students using AI in assignments

A bank of questions, varied difficulty

Topic: balancing chemical equations

Easy	**Balance: $H_2 + O_2 \rightarrow H_2O$** *Two atoms each side. One missing coefficient.*
Medium	**Balance: $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$** *Combustion of propane. Three element types to track.*
Hard	**Balance: $Fe + H_2SO_4 \rightarrow Fe_2(SO_4)_3 + H_2$** *Polyatomic ion preserved across the reaction. Watch the iron.*

Distractors that actually probe thinking

Topic: causes of World War I, grade 10 history

Which event most directly triggered the start of WWI?

A. **The assassination of Archduke Franz Ferdinand**

B. The sinking of the Lusitania

C. The signing of the Treaty of Versailles

D. Germany's invasion of Belgium

Why these distractors work

B: real WWI event, but it happened in 1915.

C: ended the war, didn't start it.

D: real and important, but came after the trigger.

Make-up versions in seconds

ORIGINAL

Quiz question, grade 9 algebra

A train leaves Geneva at 9:00 a.m. travelling at 80 km/h. Another train leaves Lausanne at 9:30 a.m. travelling at 100 km/h on the same line, 60 km behind. At what time does the second train catch up?

Variant 1

A bus leaves Bern at 8:00 a.m. at 60 km/h. A car leaves Thun at 8:15 a.m. at 90 km/h, 30 km behind. When does the car catch up?

Variant 2

Cyclist A starts at 10:00 a.m. at 20 km/h. Cyclist B starts at 10:20 a.m. at 30 km/h, 10 km behind. When does B overtake A?

Variant 3

Boat A leaves the harbour at noon at 15 km/h. Boat B leaves at 12:45 at 25 km/h, 20 km behind. When does B catch A?

See you in 30 minutes

Welcome back. Picking up where we left off.

Generative AI in plain language

Prompting for reliable results

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Students using AI in assignments

Exit tickets and misconception probes

Topic: Romeo and Juliet, themes of fate and choice

Exit ticket questions

Quick checks at the end of class.

- In one sentence, what is the role of fate in Act 3?
- Name one moment where a character could have chosen differently.
- What does the prologue tell us before the play begins?

Misconception probes

Questions designed to surface what students get wrong.

- True or false: Romeo and Juliet die because of their parents. Explain.
- Some readers say the lovers had no choice. What's the strongest argument against?
- Is Friar Laurence responsible for the ending? Defend your answer.

Patterns across the whole class

Twenty quizzes in. AI tells you what to reteach tomorrow.

ONE STUDENT'S ANSWER

Question: Solve 2x + 6 = 18. **Student answer:** x = 12 (subtracted 6 from 18, then forgot to divide).

AI's read of the whole class (20 students)

12 *students*

Forgot to divide

Subtracted but didn't divide by the coefficient.

Reteach: order of inverse operations.

5 *students*

Sign error

Added instead of subtracting when moving terms across the equation.

Reteach: balancing equations.

3 *students*

Got it right

Clean work, correct answer, all steps shown.

Stretch: harder problem set ready.

Generative AI in plain language

Prompting for reliable results

AI for classroom content

- Lesson materials
- Student practice
- Formative assessment
- **The verification habit**

Students using AI in assignments

When you ask AI for a source

A peer-reviewed study tested ChatGPT on 42 topics. It checked every citation it produced.

1 in 5

citations from ChatGPT-4 pointed to papers that don't exist.

Walters, W. H., & Wilder, E. I. (2023). Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13, 14045.

Verification is your job

AI predicts plausible text. It does not retrieve facts.

Treat every AI output as a draft.

Check anything you'd be embarrassed to be wrong about.

If the stakes are high, verify against a source you trust.

What we covered

AI for classroom content

Lesson materials: explanations, analogies, worked examples, differentiated versions.

Student practice: question banks, multiple-choice distractors, make-up exams.

Formative assessment: exit tickets, misconception probes, patterns across the whole class.

The verification habit: AI predicts plausible text. Verification is your job.

Students using AI in assignments

Generative AI in plain language

Prompting for reliable results

AI for classroom content

Students using AI in assignments

- **The reality**
- Assignment design that works with AI
- Talking to students about AI use

Detection tools are unreliable

These tools can miss students who used AI.

And they can also flag students who didn't use AI.

Both error rates are too high to act on with confidence.

Catching AI use is not a reliable foundation for assessment.

Reframe the question

From

How do we catch them?

→

To

How do we redesign assessment?

Generative AI in plain language

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Make the process visible

Process-visible work

Drafts, outlines, revisions, and source notes turned in alongside the final piece.

In-class components

Discussion, short writes, problem-solving, or presentation built on the take-home work.

The work students do becomes harder to outsource when you can see how they got there.

Oral defense and critique-the-AI

Oral defense

Five minutes per student. Two questions about the choices behind their work.

Use AI and critique it

Students prompt the AI, then mark up what it got wrong, missing, or misleading.

This is the verification habit turned into student practice.

Example: essay

What AI does well

Produce a fluent, well-structured essay on almost any topic.

Build a coherent argument and defend it in writing.

What we assess instead

Whether the student can defend the choices behind the essay.

A short in-class question on a contested choice. Why this thesis. Why this counter-example. Why this source.

Example: lab report

What AI does well

Analyze any dataset you give it, including the student's own measurements.

Write a discussion section that fits the data.

What we assess instead

Whether the student actually ran the experiment.

Photo of the setup, lab notebook entries, the failed first run, raw timestamps. Submitted alongside the analysis.

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Per-assignment policies

The rules genuinely differ by assignment.

No AI: in-class work, oral assessments, certain exams.

AI with disclosure: brainstorming, drafting, feedback.

AI as the subject: assignments where students critique its output.

Disclose when you use AI yourself

If you used AI to draft a worksheet, a reading material, or a slide deck, say so.

Students learn the norm by watching you live it.

Disclosure makes AI use a normal part of the conversation, not a secret.

It earns you the standing to ask for the same from them.

What we covered

Detection is unreliable. The job is redesign, not policing.

Process-visible work, in-class components, oral defense, and critique-the-AI assignments hold up.

Per-assignment policies, with disclosure on both sides.

Conclusion

Three takeaways

1. AI is a capable but unreliable assistant. Verification is your job.
2. Prompting is a skill that compounds with practice.
3. The goal isn't preventing AI use but redesigning assignments.

Resources

1. **These slides.**
2. **The reusable patterns from Section 2.**

Further reading

One Useful Thing by Ethan Mollick.

AI Fluency for educators by Anthropic Academy.

Built with AI

77 of 77

Slides built with AI. Ground truth, verification, and direction by me.

Questions